

VGGT: Visual Geometry Grounded Transformer

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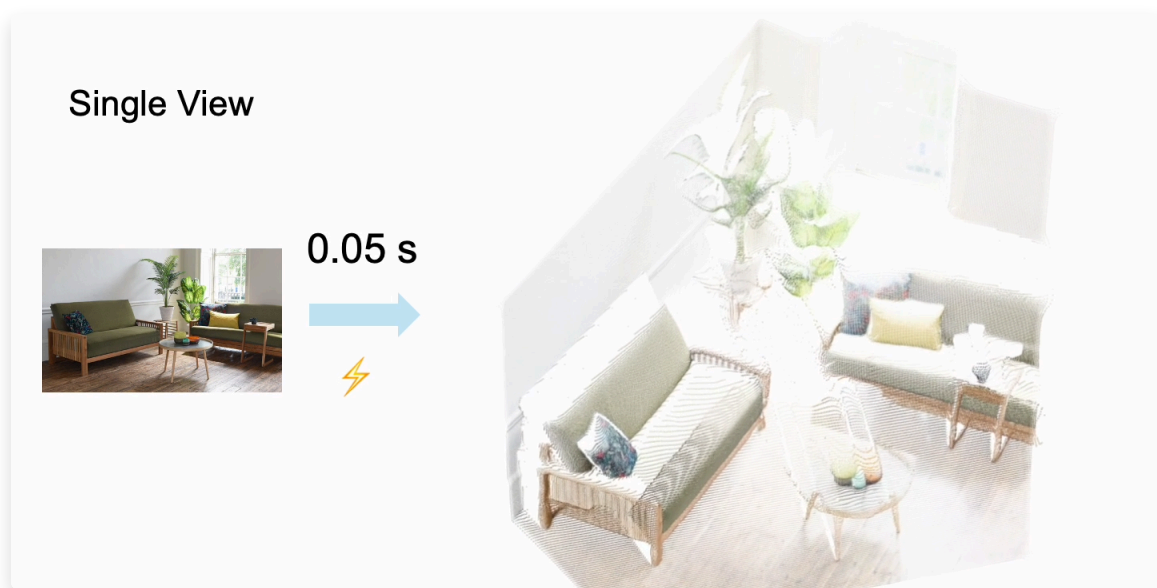
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CVPR 2025 (Oral)

[Paper](#)

[Code](#)

[👤 Demo](#)

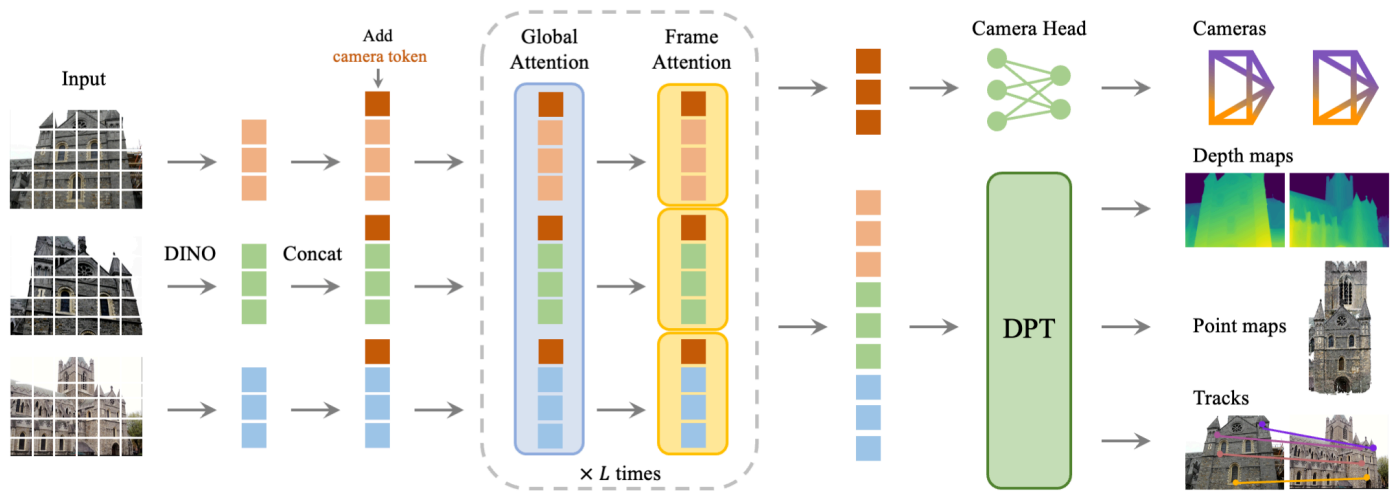


Abstract

We present **VGGT**, a feed-forward neural network that directly infers all key 3D attributes of a scene, including camera parameters, point maps, depth maps, and 3D point tracks, from one, a few, or hundreds of its views. VGGT is a step forward in 3D computer vision, where models have typically been constrained to narrow and special single tasks. It is also simple and efficient, reconstructing images in under one second, and still outperforms alternatives without their post-processing utilizing visual geometry optimization techniques. The network achieves state-of-the-art results in multiple 3D tasks, including camera parameter estimation, multi-view depth estimation, dense point cloud reconstruction, and point tracking. We also show that using pretrained **VGGT** as a feature backbone significantly enhances downstream tasks, such as non-rigid point tracking and feed-forward novel view synthesis.

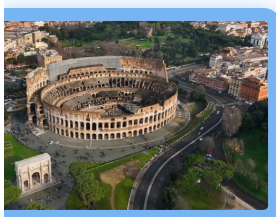
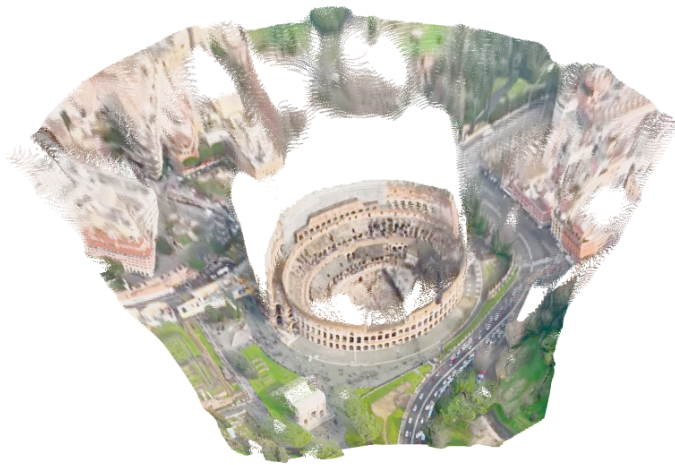
Method

VGGT first patchifies the input images into tokens by DINO, and appends camera tokens for camera prediction. It alternates between frame-wise and global self attention layers. A camera head makes the final prediction for camera extrinsics and intrinsics, while a DPT head provides any dense output, such as depth maps, point maps, or surface maps for tracking.



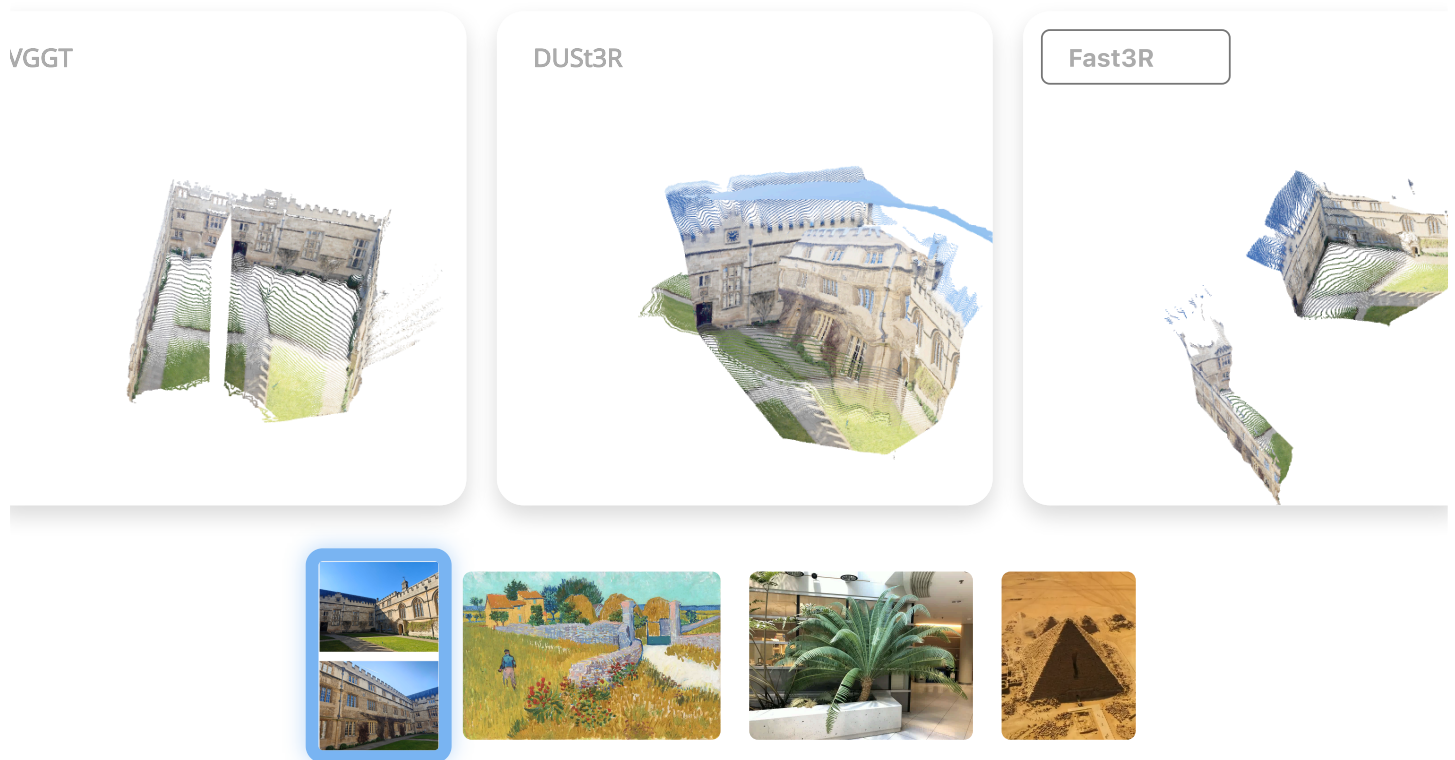
Qualitative Visualization

Reconstruction of In-the-wild Photos/Videos with VGGT. Click on any thumbnail below to view the 3D reconstruction.



Qualitative Comparison

iT significantly outperforms all other methods across various tasks. Please refer to our paper for quantitative results. Here we also provide a qualitative comparison with DUST3R and other concurrent works such as Fast3R (use the dropdown menu to switch).



BibTeX

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@inproceedings{wang2025vggt,  
  title={VGGT: Visual Geometry Grounded Transformer},  
  author={Wang, Jianyuan and Chen, Minghao and Karaev, Nikita and Vedaldi, Andrea and Ravi, Arash},  
  booktitle={Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition},  
  year={2025}  
}
```

Acknowledgements

Jianyuan Wang is supported by Facebook Research.

We are deeply grateful for the insightful discussions and invaluable support provided by Stanislaw Szymanowicz, Junyu Xie, Johannes Schönberger, Shangzhe Wu, Chuanxia Zheng, Junlin Han, Ang Cao,

Nikhil Keetha, Chris Offner, Shangzhan Zhang, Yuxi Xiao, Qianqian Wang, Yinghao Xu, Ceyuan Yang, Nan Xue, Yujun Shen, Roman Shapovalov, João Henriques, and Andrew Zisserman.

We appreciate the great examples provided by Depth-Anything-V2, Metric3D V2, MoGe, and FLARE.

Special thanks to Jianing Yang, Ang Cao, Zhenggang Tang, Yuchen Fan, Shangzhan Zhang, and Qianqian Wang for providing or verifying the results of their methods.

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